Power tree: fused 12–24 V input with a TVS and two independent 1 A series-diode feeds. One feed powers the 5 V logic buck and analog housekeeping; the other powers the nominal 11 V backlight buck. WeAct onboard 3.3 V regulation supplies only the module, and the ESP has its own 3.3 V LDO.

1=24+ 2=-

Phoenix Contact MCV 1.5/2-G-3.5

SMBJ24CA

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LMR36510ADDAR

Default: the LMR36510 generates 5V for the module and both peripheral LDOs. The WeAct onboard regulator supplies only the module, whose 3V3 outputs and VBUS remain local.

GND 5V ADC_3V3 VIN_P ENC_5V SD_3V3 PLC_24V PLC_0V BL_VIN BL_12V

J2 carries the isolated 12 V backlight rail generated from the PLC 24 V field supply and its PLC_0V return. Backlight PWM is not provided by this board.

1=BL12+ 2=PLC-

Phoenix Contact MCV 1.5/2-G-3.5

U3 is a 24 V to 12 V LMR36510 buck powered from the isolated PLC field supply on J33/J45. D3 provides reverse protection and the complete converter loop references PLC_0V.

SS16

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LMR36510ADDAR

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